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Roll No

CE-5001-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Transportation Engineering

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume suitable missing value.

1. a) Draw a typical cross-section of a permanent way. Discuss in brief the basic functions of various components of the railway track. 7
b) Discuss the necessity and effects of coning of wheels and tilting of rails. 7
2. Derive the formula of super elevation on railway track. Explain the negative super elevation. A 5° curve diverges from 3° curve in reverse direction in the layout of a BG yard, if the speed on the branch line is restricted to 35 km/h. Determine the restricted speed on main line. 14
3. a) What do you mean by “Economical span” for a bridge? Discuss the factors on which the selection of a loading types depends in case of bridge design. 7
b) Define strengthening of bridge. Explain any one method of strengthening of a bridge. 7

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4. Write short notes on: 14
- a) Shape and size of tunnel
 - b) Mucking operation
 - c) Drainage and Ventilation
 - d) Pressure relief phenomenon
5. a) Discuss advantages and disadvantages of inland waterways. 7
- b) Explain dry docks and wet docks. 7
6. Discuss each types of gradient used on railway track. What are the reasons for compensation of grade? What should be the actual ruling gradient? If the ruling gradient of 1 in 150 or a B.G. and a curve of 3° is superimposed. 14
7. a) Discuss the function of ballast on railway work. What are the requirement of good ballast? 7
- b) Discuss different types of signals used in stations and yards. 7
8. a) Explain Indian loading standards for railway bridges and highway bridges. 7
- b) What is the necessity of railway tunnels? Draw a sketch to illustrate a single track railway tunnel. 7
